

Experiment No. 24

Title

Prolog Program to Suggest Dieting System Based on Disease

Aim

To write a Prolog program to suggest a suitable **diet based on a disease**.

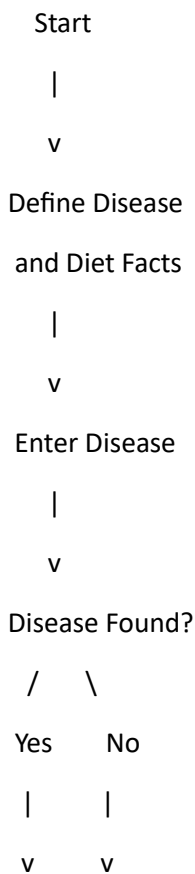
Objective

The objective of this experiment is to use Prolog facts and rules to provide diet recommendations for different diseases.

Algorithm

1. Start.
2. Define diseases and their recommended diets.
3. Enter a disease query.
4. Match the disease with the database.
5. Display the recommended diet.
6. Stop.

Flowchart



Display Diet Display

Not Found

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v v

Stop

Prolog Program

diet(diabetes, 'Low sugar and high fiber').

diet(hypertension, 'Low salt and low fat').

diet(obesity, 'Low calorie and high protein').

diet(anemia, 'Iron rich foods').

diet(cold, 'Warm fluids and fruits').

suggest_diet(Disease, Diet) :-

 diet(Disease, Diet).

Queries

?- suggest_diet(diabetes, Diet).

Output:

Diet = 'Low sugar and high fiber'.

?- suggest_diet(anemia, Diet).

Output:

Diet = 'Iron rich foods'.

Result

The Prolog program successfully suggested a suitable diet based on the given disease.

Conclusion

Thus, a simple **disease-based dieting system** was successfully implemented using Prolog facts and queries.